

Problem 18. The Lucas numbers are defined by the same recurrence relation as the Fibonacci numbers, $L_n = L_{n-1} + L_{n-2}$, with a different initial condition: $L_0 = 2$, $L_1 = 1$. Find an explicit formula for L_n in terms of n .

Solution. We can write the recurrence relation for the Lucas numbers in matrix form, as for the Fibonacci numbers:

$$\begin{bmatrix} L_n \\ L_{n+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} L_{n-1} \\ L_n \end{bmatrix}$$

Call the matrix A . We then have

$$\begin{bmatrix} L_n \\ L_{n+1} \end{bmatrix} = A^n \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

To turn this into an explicit formula for L_n , we will diagonalize A . This is the same as in the case of the Fibonacci sequence, but I will reproduce it here.

The characteristic polynomial is

$$\det \begin{bmatrix} -t & 1 \\ 1 & 1-t \end{bmatrix} = t^2 - t - 1$$

which has roots

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}$$

We have $\varphi\psi = -1$ and so $\psi = -\varphi^{-1}$. Also, $\varphi + \psi = 1$. The eigenspaces are as follows:

$$V_\varphi = \ker \begin{bmatrix} -\varphi & 1 \\ 1 & 1-\varphi \end{bmatrix} = \ker \begin{bmatrix} -\varphi & 1 \\ 1 & -\varphi^{-1} \end{bmatrix} = \text{span} \left\{ \begin{bmatrix} 1 \\ \varphi \end{bmatrix} \right\}$$

$$V_\psi = \ker \begin{bmatrix} -\psi & 1 \\ 1 & 1-\psi \end{bmatrix} = \ker \begin{bmatrix} -\psi & 1 \\ 1 & -\psi^{-1} \end{bmatrix} = \text{span} \left\{ \begin{bmatrix} 1 \\ \psi \end{bmatrix} \right\}$$

Thus we have diagonalization

$$A = \begin{bmatrix} 1 & 1 \\ \varphi & \psi \end{bmatrix} \begin{bmatrix} \varphi & 0 \\ 0 & \psi \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \varphi & \psi \end{bmatrix}^{-1}$$

Using our general formula for the inverse of a 2×2 matrix, we have

$$\begin{bmatrix} 1 & 1 \\ \varphi & \psi \end{bmatrix}^{-1} = \frac{1}{\psi - \varphi} \begin{bmatrix} \psi & -1 \\ -\varphi & 1 \end{bmatrix} = \frac{1}{-\sqrt{5}} \begin{bmatrix} \psi & -1 \\ -\varphi & 1 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} -\psi & 1 \\ \varphi & -1 \end{bmatrix}$$

We then get the following formula:

$$\begin{bmatrix} L_n \\ L_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \varphi & \psi \end{bmatrix} \begin{bmatrix} \varphi^n & 0 \\ 0 & \psi^n \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} -\psi & 1 \\ \varphi & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} \varphi^n & \psi^n \\ \varphi^{n+1} & \psi^{n+1} \end{bmatrix} \begin{bmatrix} 1 - 2\psi \\ 2\varphi - 1 \end{bmatrix}$$

Looking at the first entry of this product, we get

$$L_n = \frac{\varphi^n - 2\psi\varphi^n + 2\varphi\psi^n - \psi^n}{\sqrt{5}} = \frac{\varphi^n + 2\varphi^{n-1} - 2\psi^{n-1} - \psi^n}{\sqrt{5}}$$

You can check that indeed

$$L_0 = \frac{\varphi^0 - 2\psi + 2\varphi - \psi^0}{\sqrt{5}} = \frac{2\sqrt{5}}{\sqrt{5}} = 2$$

and

$$L_1 = \frac{\varphi + 2 - 2 - \psi}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1.$$